APPENDIX I. VANUATU: PUBLIC AND EXTERNAL DEBT SUSTAINABILITY[1,2]

This appendix summarizes the debt sustainability analysis (DSA) for Vanuatu using the framework for low-income countries. Vanuatu's public debt has been declining on average, reflecting strong growth performance, successful fiscal consolidation, and large aid inflows which eschewed the need for borrowing. As a result, most debt burden indicators under the baseline scenario are below their indicative thresholds. This analysis suggests that Vanuatu's public debt dynamics are sustainable and the country faces a low level of debt distress.

I. INTRODUCTION AND BASELINE

Reflecting strong growth and fiscal surpluses over the last few years, Vanuatu's total public sector debt stock is estimated to have declined to 18½ percent of GDP in 2008 from 28 percent of GDP in 2005. The ratio of external debt to GDP fell from over 30 percent in 2000 to 13¾ percent in 2007, inching up a little to 15¾ percent in 2008, due to a new bilateral loan. This DSA presents the projected path of Vanuatu's external and public sector debt burden indicators, and draws some conclusions on the forward-looking sustainability of debt.

Vanuatu's DSA builds on the baseline scenario assumptions presented in Box I.1. It assumes real GDP growth rate of about 4½ percent on average over the medium term, inflation of about 3 percent, and modest fiscal surplus. The baseline assumes no additional external borrowing other than the new bilateral loan contracted in 2008.[3] The authorities are strongly committed to fiscal consolidation and do not anticipate the need to borrow given donor flows.

II. EXTERNAL DEBT SUSTAINABILITY

External debt stood at 15¾ percent of GDP in FY2008, down from over 20 percent in FY2005 (Table I.1a, Figure I.1). The government took on a new loan in 2008 to expedite improvements in IT-related infrastructure. Under the baseline scenario, the NPV of external debt is projected at 10½ percent of GDP in 2008 (below the indicative threshold based on the CPIA rating which assumes Vanuatu to be a weak reformer).[4] This ratio is expected to fall to

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[1] In light of the low debt-to-GDP ratios, this DSA was prepared by Fund staff only.

[2] Data on state enterprise and provincial government debt are not available. Therefore, the public debt sustainability analysis is limited to central government debt data.

[3] It is a bilateral loan of about US$27 million at 2 percent interest, with a 5-year grace period and 20-year maturity.

[4] The average World Bank's Country Policy and Institutional Assessment for the period 2005–07 classifies Vanuatu as a poor performer. The corresponding debt distress thresholds are NPV of debt-to-exports (100 percent), GDP (30 percent), and revenues (200 percent), and debt service in percent of exports (15 percent) and revenues (25 percent).

9¼ percent by FY2013 and further in the medium term. The NPV of debt-to-exports and revenue ratios increase somewhat in 2009–10, but declining thereafter. All the ratios (including PPG debt service in percent of exports and revenue) remain well below their respective indicative thresholds as shown.

The alternative scenarios and bound tests indicate that external debt sustainability is sensitive to an exchange rate shock (Table I.1b, Figure I.1). If growth is assumed to be at its historical average, the NPV of debt-to-GDP ratio falls, reflecting the strong performance of the last five years. Among the various stress tests, bound test 6 (30 percent nominal depreciation of the exchange rate) results in the largest increase in the NPV of debt-to-GDP ratio, which jumps to 18 percent in 2009–10, then declines gradually. None of the stress tests cause any breaches in the thresholds.

III. Public Debt Sustainability

Vanuatu's public debt has declined from over 40 percent of GDP in 2002 to 18½ percent of GDP in 2008, reflecting the authorities' commitment to fiscal consolidation (Table I.2a). The NPV of debt-to-GDP is expected to decline from 14½ percent at end-2008, to 10¾ percent in 2013, and further to around 1 percent by end-2028.

Public debt is sensitive to exchange rate changes, given the dominance of external debt in total public debt. Compared to 2008, a one-time 30 percent real depreciation of the exchange rate would raise the NPV of public debt-to-GDP by 6 percent age points in 2009, mirroring the impact on external debt, and fall gradually thereafter. A sizable growth shock in 2009–10 (economy is assumed to contract by 11½ percent in two years compared to the baseline) will likely have a lasting effect over the medium term. The baseline scenario is more favorable than the historical one, reflecting the success of government's fiscal consolidation efforts and commitment to continue this in the future. The other factors have a modest impact on the NPV of debt-to-GDP.

IV. Staff Assessment

Assuming Vanuatu continues to make steady progress in reducing its public debt burden, as it has done in the past few years, public debt sustainability should be manageable. While several indicators show that the economy's debt is low, debt remains somewhat vulnerable to a major shock in the exchange rate. Also, changes in GDP growth in particular affect the debt dynamics. Given that both growth and balance of payments are driven by tourism, the external environment, especially the current global environment does pose a challenge. Therefore, the authorities must remain committed to bringing down the external debt further to reduce vulnerabilities. Going forward, improved coordination with donors to align their plans with authorities' development objectives can help to mobilize/use aid inflows more effectively.

Box I.1. Assumptions for the Medium-Term Baseline Scenario

Real GDP growth is projected to average 4½ percent over the medium term as further structural reforms, including improvements in infrastructure, boost productivity. After moderating in 2009–10, tourism is expected to grow at a somewhat more moderate pace than in the last few years.

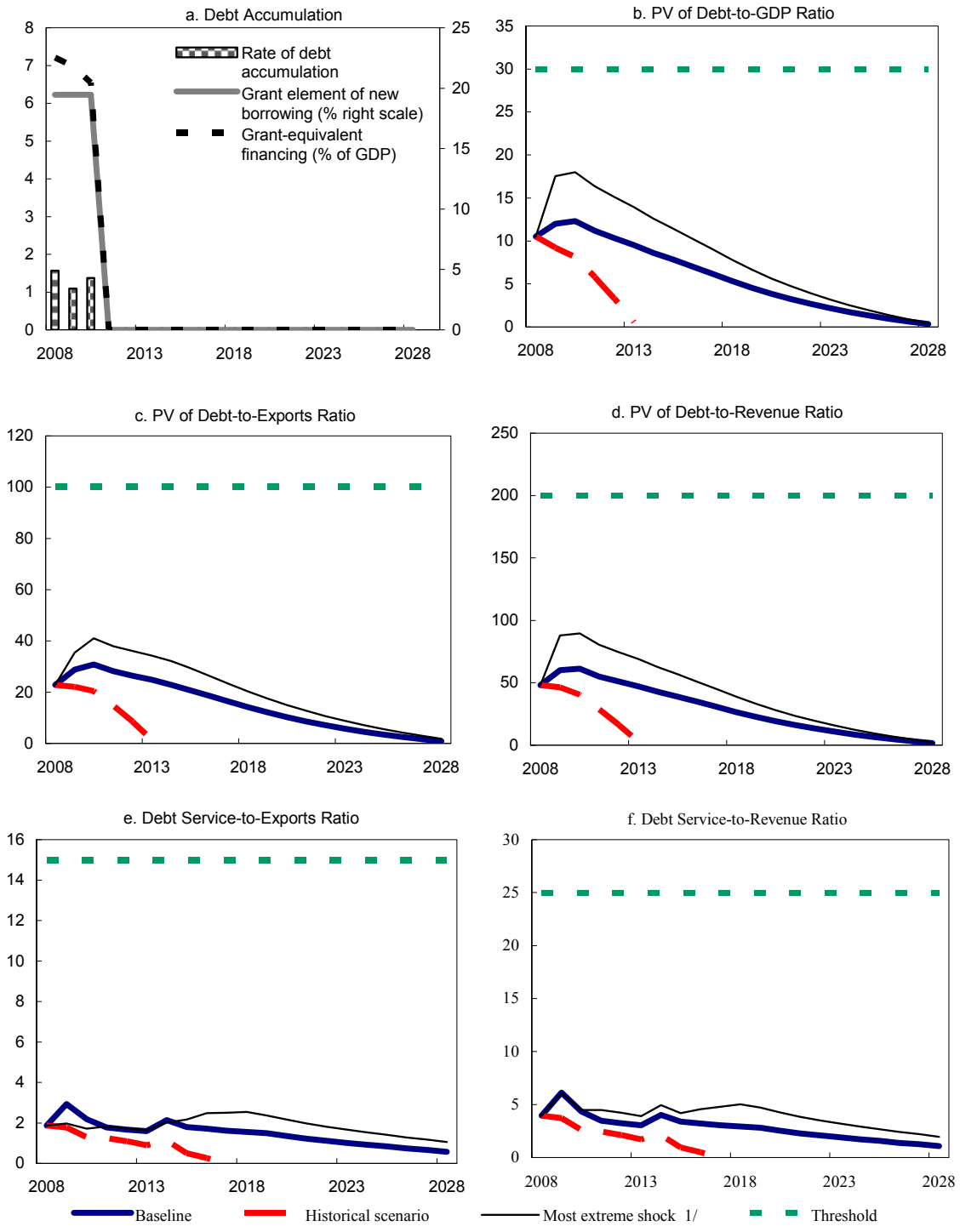
Inflation is projected at around 3 percent over the projection period. With a transaction-based exchange regime providing an effective inflation anchor, inflation has been on average 2½ percent in the recent five years.

The overall fiscal balance is projected to register an average surplus of ¼ percent of GDP over the medium term given the recent strong fiscal performances supported by authorities' commitment to a sound fiscal policy. This envisages a modest fiscal expansion (in the current expenditure) in the near term, but assumes that the authorities limit expenditures as a share of GDP after 2012.

The current account deficit is expected to decline in 2009−10 as import prices are expected to fall (around 5 percent of GDP) while economic activity will slow with some lags from the global economy, but is expected to widen to about 6½ percent by 2014, due mainly to a substantial pick up in tourism-related imports financed by FDI inflows. The real effective exchange rate is assumed constant.

Aid inflows are assumed to remain stable. FDI related to tourism and real estate is expected to increase significantly. With these developments, reserve cover is projected to remain around six months of projected imports by 2014.

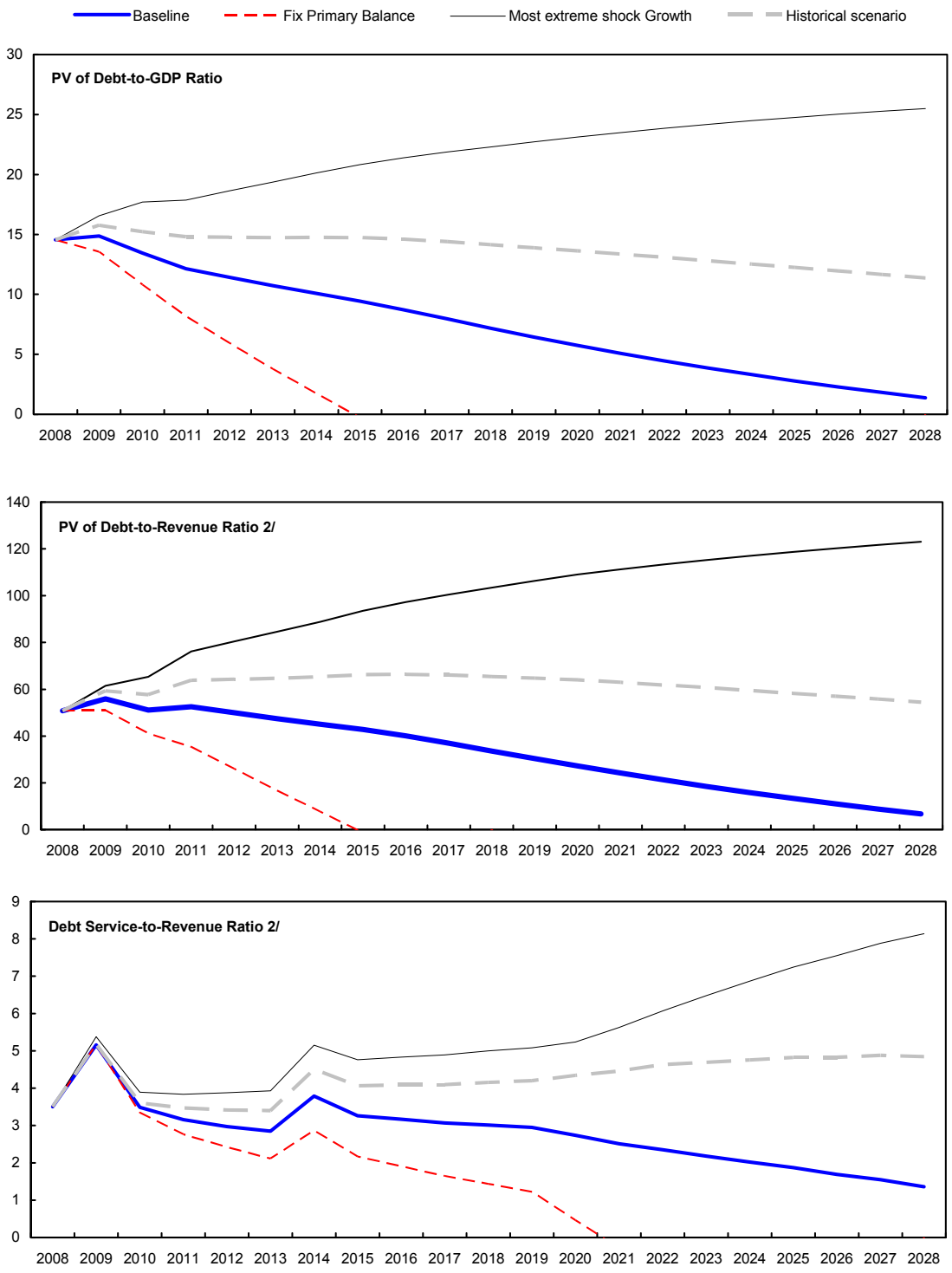
Figure I.1. Vanuatu: Indicators of Public and Publicly Guaranteed External Debt under Alternative Scenarios, 2008–28 1/



Source: IMF staff projections and simulations.

1/ The most extreme stress test is the test that yields the highest ratio in 2018. In figure b. it corresponds to a nondebt flows shock; in c. to a nondebt flows shock; in d. to a nondebt flows shock; in e. to a nondebt flows shock and in picture figure f. to a nondebt flows shock.

Figure I.2. Vanuatu: Indicators for Public Debt under Alternative Scenarios, 2008–28 1/



Sources: Country authorities; and IMF staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in 2018.

2/ Revenues are defined inclusive of grants.

Table I.1a. Vanuatu: External Debt Sustainability Framework, Baseline Scenario, 2005–28 1/
(In percent of GDP, unless otherwise indicated)

	Actual			Historical Average 2/	Standard Deviation 2/	Est.	Projections								
	2005	2006	2007			2008	2009	2010	2011	2012	2013	2008–2013 Average	2018	2028	2014–2028 Average
External debt (nominal) 1/	**20.4**	**16.1**	**13.7**			**15.7**	**14.4**	**14.4**	**13.2**	**11.9**	**10.6**		**5.3**	**0.5**	
Of which: public and publicly guranteed (PPG)	20.4	16.1	13.7			15.7	14.4	14.4	13.2	11.9	10.6		5.3	0.5	
Change in external debt	-2.7	-4.3	-2.4			1.9	-1.3	0.0	-1.2	-1.4	-1.2		-0.9	-0.2	
Identified net debt-creating flows	-1.0	-5.9	-3.6			0.7	0.7	0.7	0.4	0.8	1.2		3.4	6.2	
Non-interest current account deficit	**6.8**	**3.6**	**5.5**	**3.3**	**3.5**	**6.9**	**5.3**	**4.9**	**5.3**	**5.6**	**6.0**		**6.9**	**7.8**	7.2
Deficit in balance of goods and services	5.8	5.3	5.6			12.4	12.0	10.0	8.3	8.4	8.6		8.6	8.6	
Exports	47.9	44.1	42.3			46.0	41.7	39.9	39.7	39.0	38.4		37.8	37.8	
Imports	53.7	49.5	47.9			58.4	53.6	49.9	47.9	47.4	47.0		46.4	46.4	
Net current transfers (negative = inflow)	-5.4	-6.1	-5.1	-4.4	2.3	-7.3	-6.7	-5.4	-3.5	-3.2	-3.0		-2.1	-1.0	-1.7
Of which: official	0.0	-0.8	-0.2			-0.6	-0.5	-0.5	-0.4	-0.4	-0.4		0.0	0.0	
Other current account flows (negative = net inflow)	6.4	4.4	5.0			1.8	0.0	0.3	0.4	0.5	0.5		0.3	0.2	
Net FDI (negative = inflow)	**-5.9**	**-7.8**	**-6.6**	**-6.7**	**1.1**	**-5.8**	**-4.9**	**-4.2**	**-4.6**	**-4.5**	**-4.5**		**-3.4**	**-1.6**	-2.8
Endogenous debt dynamics 3/	**-1.9**	**-1.7**	**-2.5**			**-0.4**	**0.3**	**0.0**	**-0.3**	**-0.3**	**-0.3**		**-0.1**	**0.0**	
Contribution from nominal interest rate	0.6	0.5	0.4			0.4	0.8	0.4	0.3	0.2	0.2		0.2	0.0	
Contribution from real GDP growth	-1.3	-1.3	-0.9			-0.8	-0.5	-0.5	-0.6	-0.6	-0.5		-0.3	0.0	
Contribution from price and exchange rate changes	-1.2	-0.9	-2.0			…	…	…	…	…	…		…	…	
Residual (3–4) 4/	**-1.7**	**1.6**	**1.2**			**1.2**	**-2.0**	**-0.7**	**-1.6**	**-2.1**	**-2.5**		**-4.3**	**-6.4**	
Of which: exceptional financing	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt 5/	...	...	10.3			10.5	12.0	12.3	11.2	10.3	9.5		5.3	0.3	
In percent of exports	...	...	24.3			22.8	28.8	30.8	28.2	26.4	24.8		14.1	0.9	
PV of PPG external debt	**...**	**...**	**10.3**			**10.5**	**12.0**	**12.3**	**11.2**	**10.3**	**9.5**		**5.3**	**0.3**	
In percent of exports	**...**	**...**	**24.3**			**22.8**	**28.8**	**30.8**	**28.2**	**26.4**	**24.8**		**14.1**	**0.9**	
In percent of government revenues	**...**	**...**	**48.4**			**48.2**	**60.2**	**61.2**	**55.0**	**51.0**	**47.2**		**26.5**	**1.6**	
Debt service-to-exports ratio (in percent)	**4.0**	**2.1**	**1.9**			**1.9**	**2.9**	**2.2**	**1.8**	**1.7**	**1.6**		**1.6**	**0.6**	
PPG debt service-to-exports ratio (in percent)	**4.0**	**2.1**	**1.9**			**1.9**	**2.9**	**2.2**	**1.8**	**1.7**	**1.6**		**1.6**	**0.6**	
PPG debt service-to-revenue ratio (in percent)	**9.5**	**4.7**	**3.8**			**3.9**	**6.1**	**4.4**	**3.5**	**3.2**	**3.1**		**2.9**	**1.1**	
Total gross financing need (Billions of U.S. dollars)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.1	
Non-interest current account deficit that stabilizes debt ratio	9.5	7.9	7.8			5.0	6.6	4.9	6.4	7.0	7.3		7.8	8.1	
Key macroeconomic assumptions															
Real GDP growth (in percent)	6.5	7.4	6.8	2.3	5.0	6.6	3.0	3.5	4.2	4.5	4.5	4.4	4.5	4.5	4.5
GDP deflator in US dollar terms (change in percent)	5.4	4.7	14.2	5.0	7.8	6.0	-6.2	5.0	5.0	3.0	3.0	2.6	3.0	3.0	3.0
Effective interest rate (percent) 6/	2.8	3.0	3.2	3.0	0.2	3.7	5.1	3.2	2.0	1.9	2.0	3.0	2.7	4.7	3.3
Growth of exports of G&S (US dollar terms, in percent)	10.7	3.6	17.0	5.5	13.1	22.7	-12.3	4.0	8.8	5.9	5.9	5.8	7.6	7.6	7.5
Growth of imports of G&S (US dollar terms, in percent)	15.3	3.6	18.2	6.9	12.0	37.7	-11.2	1.1	5.1	6.5	6.6	7.6	7.6	7.6	7.5
Grant element of new public sector borrowing (in percent) 7/	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
Government revenues (excluding grants, in percent of GDP) 8/	19.9	19.9	21.3			21.8	19.9	20.1	20.3	20.3	20.2		20.1	20.1	20.1
Memorandum items:															
Nominal GDP (Billions of U.S. dollars)	0.4	0.4	0.5			0.6	0.6	0.6	0.7	0.7	0.8		1.1	2.3	

Source: IMF staff simulations.

1/ Due to data constraints, it includes only public external debt.
2/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.
3/ Derived as [r - g - r(1+g)]/(1+g+r+gr) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and r = growth rate of GDP deflator in U.S. dollar terms.
4/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.
5/ Assumes that PV of private sector debt is equivalent to its face value.
6/ Current-year interest payments divided by previous period debt stock.
7/ Defined as grants, concessional loans, and debt relief.
8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table I.1b. Vanuatu: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2008–28
(In percent)

	Projections											
	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2028
PV of Debt-to-GDP Ratio												
Baseline	10	12	12	11	10	10	9	8	7	6	5	0
A. Alternative Scenarios												
A1. Key variables at their historical averages in 2008–28 1/	10	9	8	6	3	1	...	...	...	...	...	...
A2. New public sector loans on less favorable terms in 2008–28 2/	10	12	13	11	11	10	9	8	7	7	6	1
B. Bound Tests												
B1. Real GDP growth at historical average minus one standard deviation in 2009–10	10	13	14	13	12	11	10	9	8	7	6	0
B2. Export value growth at historical average minus one standard deviation in 2009–10 3/	10	10	13	11	11	10	9	8	7	6	6	0
B3. U.S. dollar GDP deflator at historical average minus one standard deviation in 2009–10	10	12	13	12	11	10	9	8	7	6	6	0
B4. Net nondebt creating flows at historical average minus one standard deviation in 2009–10 4/	10	15	16	15	14	13	12	11	10	9	8	1
B5. Combination of B1-B4 using one-half standard deviation shocks	10	9	8	7	7	6	5	5	4	4	3	0
B6. One-time 30 percent nominal depreciation relative to the baseline in 2009 5/	10	18	18	16	15	14	13	11	10	9	8	0
PV of Debt-to-Exports Ratio												
Baseline	23	29	31	28	26	25	23	21	19	16	14	1
A. Alternative Scenarios												
A1. Key variables at their historical averages in 2008–28 1/	23	22	20	15	8	1	...	...	...	...	...	...
A2. New public sector loans on less favorable terms in 2008–28 2/	23	29	32	29	27	26	24	22	20	18	16	3
B. Bound Tests												
B1. Real GDP growth at historical average minus one standard deviation in 2009–10	23	29	31	28	26	25	23	21	18	16	14	1
B2. Export value growth at historical average minus one standard deviation in 2009–10 3/	23	23	34	31	29	27	25	23	21	18	16	1
B3. U.S. dollar GDP deflator at historical average minus one standard deviation in 2009–10	23	29	31	28	26	25	23	21	18	16	14	1
B4. Net nondebt creating flows at historical average minus one standard deviation in 2009–10 4/	23	35	41	38	36	34	32	29	26	23	20	2
B5. Combination of B1-B4 using one-half standard deviation shocks	23	20	19	17	15	14	12	11	10	8	7	0
B6. One-time 30 percent nominal depreciation relative to the baseline in 2009 5/	23	29	31	28	26	25	23	21	18	16	14	1
PV of Debt-to-Revenue Ratio												
Baseline	48	60	61	55	51	47	43	39	35	31	27	2
A. Alternative Scenarios												
A1. Key variables at their historical averages in 2008–28 1/	48	46	40	29	16	3	...	...	...	...	...	...
A2. New public sector loans on less favorable terms in 2008–28 2/	48	61	63	57	53	49	44	41	37	33	30	5
B. Bound Tests												
B1. Real GDP growth at historical average minus one standard deviation in 2009–10	48	63	69	62	57	53	48	44	39	34	30	2
B2. Export value growth at historical average minus one standard deviation in 2009–10 3/	48	52	62	56	52	48	44	40	36	32	28	2
B3. U.S. dollar GDP deflator at historical average minus one standard deviation in 2009–10	48	58	64	57	53	49	44	40	36	32	28	2
B4. Net nondebt creating flows at historical average minus one standard deviation in 2009–10 4/	48	74	81	74	70	65	60	55	49	44	38	3
B5. Combination of B1-B4 using one-half standard deviation shocks	48	45	41	36	32	29	25	22	20	17	15	0
B6. One-time 30 percent nominal depreciation relative to the baseline in 2009 5/	48	88	90	81	75	69	63	57	51	45	39	2

Source: IMF staff projections and simulations.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and nondebt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table I.2a. Vanuatu: Public Sector Debt Sustainability Framework, Baseline Scenario, 2005–28
(In percent of GDP, unless otherwise indicated)

	Actual					Est.	Projections								
	2005	2006	2007	Average 2/	Standard Deviation 2/	2008	2009	2010	2011	2012	2013	2008–13 Average	2018	2028	2014–28 Average
Public sector debt 1/	28.1	23.4	19.8			18.5	17.3	15.8	14.2	13.0	11.8		7.1	1.5	
Of which: foreign-currency denominated	20.4	16.1	13.7			15.7	14.4	14.4	13.2	11.9	10.6		5.3	0.5	
Change in public sector debt	-5.6	-4.6	-3.6			-1.3	-1.3	-1.5	-1.6	-1.2	-1.2		-0.8	-0.4	
Identified debt-creating flows	-3.7	-4.9	-3.9			-2.7	-1.3	-2.0	-1.6	-1.2	-1.2		-0.8	-0.4	
Primary deficit	-2.8	-1.2	-0.9	1.2	3.3	-2.9	-0.9	-0.9	-0.8	-0.5	-0.5	-1.1	-0.5	-0.4	-0.4
Revenue and grants	21.7	21.8	23.1			28.6	26.6	26.3	23.1	22.9	22.6		21.4	20.7	
Of which: grants	1.7	1.9	1.9			6.8	6.6	6.2	2.8	2.6	2.4		1.3	0.5	
Primary (noninterest) expenditure	18.8	20.5	22.2			25.8	25.7	25.5	22.3	22.4	22.1		20.9	20.3	
Automatic debt dynamics	-0.9	-3.6	-3.0			0.1	-0.4	-1.1	-0.8	-0.7	-0.7		-0.4	0.0	
Contribution from interest rate/growth differential	-2.1	-2.2	-1.6			-1.0	0.1	-0.3	-0.5	-0.5	-0.4		-0.2	0.0	
Of which: contribution from average real interest rate	-0.1	-0.2	-0.1			0.3	0.6	0.3	0.2	0.1	0.1		0.1	0.1	
Of which: contribution from real GDP growth	-2.1	-1.9	-1.5			-1.2	-0.5	-0.6	-0.6	-0.6	-0.6		-0.3	-0.1	
Contribution from real exchange rate depreciation	1.3	-1.5	-1.4			1.1	-0.5	-0.8	-0.3	-0.3	-0.2		...	...	
Other identified debt-creating flows	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g., bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	-1.9	0.3	0.3			1.4	0.1	0.5	0.0	0.0	0.0		0.0	0.0	
Other Sustainability Indicators															
PV of public sector debt	...	...	16.1			14.5	14.9	13.5	12.1	11.4	10.7		7.2	1.4	
Of which: foreign-currency denominated	...	...	10.0			11.7	12.0	12.1	11.2	10.3	9.5		5.3	0.3	
Of which: external	...	...	10.0			11.7	12.0	12.1	11.2	10.3	9.5		5.3	0.3	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 3/	...	...	6.1			3.3	4.6	2.8	1.0	1.1	1.2		1.8	1.0	
PV of public sector debt-to-revenue and grants ratio (in percent)	…	…	69.6			50.8	55.9	51.1	52.5	50.0	47.4		33.6	6.7	
PV of public sector debt-to-revenue ratio (in percent)	…	…	75.8			66.8	74.6	67.0	59.8	56.4	53.1		35.7	6.8	
Of which: external 4/	…	…	47.2			53.6	60.2	60.1	55.0	51.0	47.2		26.5	1.6	
Debt service-to-revenue and grants ratio (in percent) 5/	…	5.3	4.3			3.5	5.2	3.5	3.2	3.0	2.9		3.0	1.4	
Debt service-to-revenue ratio (in percent) 5/	…	5.8	4.7			4.6	6.9	4.6	3.6	3.4	3.2		3.2	1.4	
Primary deficit that stabilizes the debt-to-GDP ratio	2.7	3.4	2.7			-1.6	0.4	0.6	0.8	0.7	0.7		0.4	0.0	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	6.5	7.4	6.8	2.3	5.0	6.6	3.0	3.5	4.2	4.5	4.5	4.4	4.5	4.5	4.5
Average nominal interest rate on forex debt (in percent)	2.8	3.0	3.2	3.0	0.2	3.7	5.1	3.2	2.0	1.9	2.0	3.0	2.7	4.7	3.3
Average real interest rate on domestic debt (in percent)	0.1	-2.8	-2.7	-1.3	1.3	-1.9	-0.7	-1.7	-1.1	-0.5	-0.5	-1.1	0.2	2.9	1.0
Real exchange rate depreciation (in percent, + indicates depreciation)	5.8	-7.7	-9.2	-3.5	9.4	8.5	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	2.9	5.9	5.9	3.5	1.7	4.8	4.3	3.0	3.0	3.0	3.0	3.5	3.0	3.0	3.0
Growth of real primary spending (deflated by GDP deflator, in percent)	0.0	0.2	0.2	0.0	0.1	0.2	0.0	0.0	-0.1	0.0	0.0	0.0	0.0	0.0	0.0
Grant element of new external borrowing (in percent)	...	...	...	…	…	19.5	19.5	19.5	...	...	...	...	...	...	...

Sources: Country authorities; and IMF staff estimates and projections.

1/ Includes central government debt.
2/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.
3/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.
4/ Revenues excluding grants.
5/ Debt service is defined as the sum of interst and amortization of medium and long-term debt.

Table I.2b. Vanuatu: Sensitivity Analysis for Key Indicators of Public Debt, 2008–28

	Projections							
	2008	2009	2010	2011	2012	2013	2018	2028
PV of Debt-to-GDP Ratio								
Baseline	15	15	13	12	11	11	7	1
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	15	16	15	15	15	15	14	11
A2. Primary balance is unchanged from 2008	15	14	11	8	6	4	...	...
A3. Permanently lower GDP growth 1/	15	15	14	13	13	14	16	31
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2009-2010	15	17	18	18	19	19	22	26
B2. Primary balance is at historical average minus one standard deviations in 2009-2010	15	18	19	17	17	16	11	4
B3. Combination of B1-B2 using one half standard deviation shocks	15	17	18	17	17	17	17	15
B4. One-time 30 percent real depreciation in 2009	15	21	19	17	16	15	11	3
B5. 10 percent of GDP increase in other debt-creating flows in 2009	15	22	20	18	18	17	12	4
PV of Debt-to-Revenue Ratio 2/								
Baseline	51	56	51	53	50	47	34	7
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	51	59	58	64	64	65	65	54
A2. Primary balance is unchanged from 2008	51	51	41	35	26	17	...	...
A3. Permanently lower GDP growth 1/	51	57	54	58	59	60	75	150
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2009-2010	51	61	65	76	80	85	103	123
B2. Primary balance is at historical average minus one standard deviations in 2009-2010	51	67	72	76	73	69	53	18
B3. Combination of B1-B2 using one half standard deviation shocks	51	64	67	75	76	77	79	74
B4. One-time 30 percent real depreciation in 2009	51	77	71	74	71	68	51	16
B5. 10 percent of GDP increase in other debt-creating flows in 2009	51	82	76	80	77	73	57	20
Debt Service-to-Revenue Ratio 2/								
Baseline	4	5	3	3	3	3	3	1
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	4	5	4	3	3	3	4	5
A2. Primary balance is unchanged from 2008	4	5	3	3	2	2	1	...
A3. Permanently lower GDP growth 1/	4	5	4	3	3	3	4	7
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2009-2010	4	5	4	4	4	4	5	8
B2. Primary balance is at historical average minus one standard deviations in 2009-2010	4	5	4	4	4	3	3	3
B3. Combination of B1-B2 using one half standard deviation shocks	4	5	4	4	4	4	4	6
B4. One-time 30 percent real depreciation in 2009	4	6	5	5	4	4	4	3
B5. 10 percent of GDP increase in other debt-creating flows in 2009	4	5	4	4	4	4	4	3

Sources: Country authorities; and IMF staff estimates and projections.

1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.

2/ Revenues are defined inclusive of grants.